

Spotify and Weather Correlation



Sara Subedi, Eric Huber, Oliver Brown

Problem Definition & Motivation

Problem Definition:

- Investigating the correlation between weather conditions and the mood of music.
- Using Spotify's happiness score.
- Understand how environmental factors influence musical preferences and emotional states.

Motivation:

- Music significantly affects emotions, and its mood can be quantified (e.g., happiness score).
- Weather impacts human emotions, it may also affect music listening preferences.
- Motivated by the hypothesis that weather conditions influence the mood of music people listen to.

Objectives:

- To collect and analyze Spotify data to quantify the mood of music through the happiness score.
- To examine weather data during the same periods of time to identify prevalent weather conditions.
- To conduct a statistical analysis to determine if there is a significant correlation between weather conditions and the mood of music (through happiness score).

Dataset Description

Spotify

- danceability
- energy
- key
- loudness
- mode
- speechiness
- acousticness
- instrumentalness
- liveness
- **valence - “describes the musical positiveness conveyed by a track”**
- tempo

Weather

- Location Info
 - name, region, country, lat, lon, tz_id, localtime_epoch, localtime
- Forecast Info:
 - **date**, date_epoch
- By day:
 - **maxtemp_c**, maxtemp_f, **mintemp_c**, mintemp_f, **avgtemp_c**, avgtemp_f, maxwind_mph, **maxwind_kph**, **totalprecip_mm**, totalprecip_in, totalsnow_cm, **avgvis_km**, avgvis_miles, avghumidity, daily_will_it_rain, daily_chance_of_rain, daily_will_it_snow, daily_chance_of_snow, text, icon, code, **uv**
- Astronomical information:
 - sunrise, sunset, moonrise, moonset, moon_phase, moon_illumination
- By hour
 - time_epoch, time, temp_c, temp_f, is_day, condition (text, icon, code), wind_mph, wind_kph, wind_degree, wind_dir, pressure_mb, pressure_in, precip_mm, precip_in, snow_cm, humidity, cloud, feelslike_c, feelslike_f, windchill_c, windchill_f, heatindex_c, heatindex_f, dewpoint_c, dewpoint_f, will_it_rain, chance_of_rain, will_it_snow, chance_of_snow, vis_km, vis_miles, gust_mph, gust_kph, uv

Dataset Description

Spotify

- danceability
- energy
- key
- loudness
- mode
- speechiness
- acousticness
- instrumentalness
- liveness
- **valence** - “describes the musical positiveness conveyed by a track”
- tempo

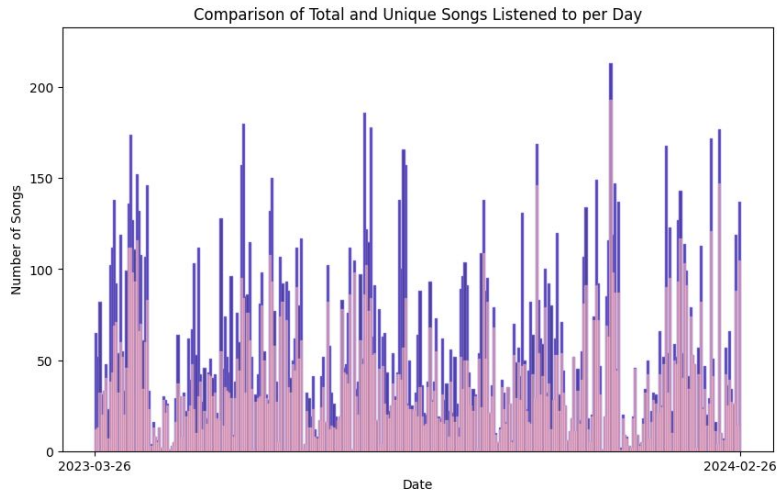
Weather

- **Date**
- **Maximum Temperature (maxtemp_c)**
- **Minimum Temperature (mintemp_c)**
- **Average Temperature (avgtemp_c)**
- **Total Precipitation (totalprecip_mm)**
- **Maximum Wind Speed (maxwind_mph)**
- **Average Visibility (avgvis_km)**
- **Average Humidity (avghumidity)**
- **UV Index (uv):**

Methodology

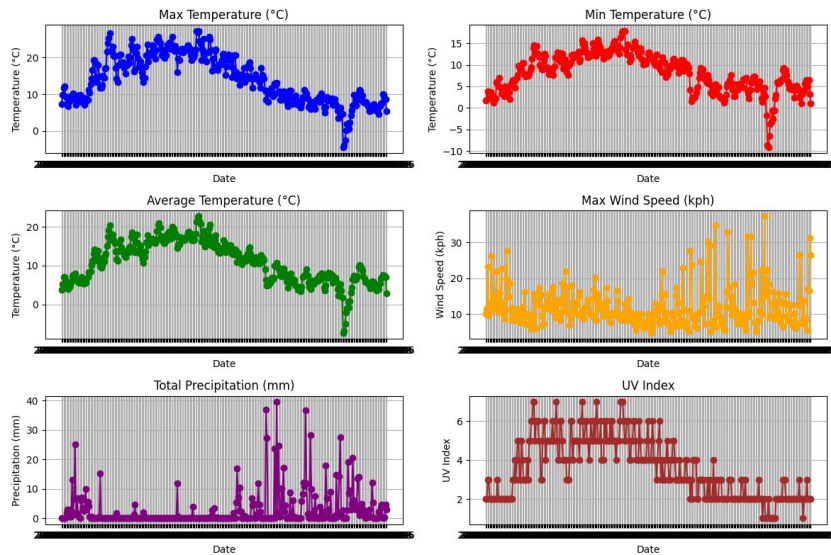
Spotify

- Used personal streaming history
- From March 26, 2023 to February 26, 2024 (due to weather API restriction)
- Any uri that returned nothing were ignored

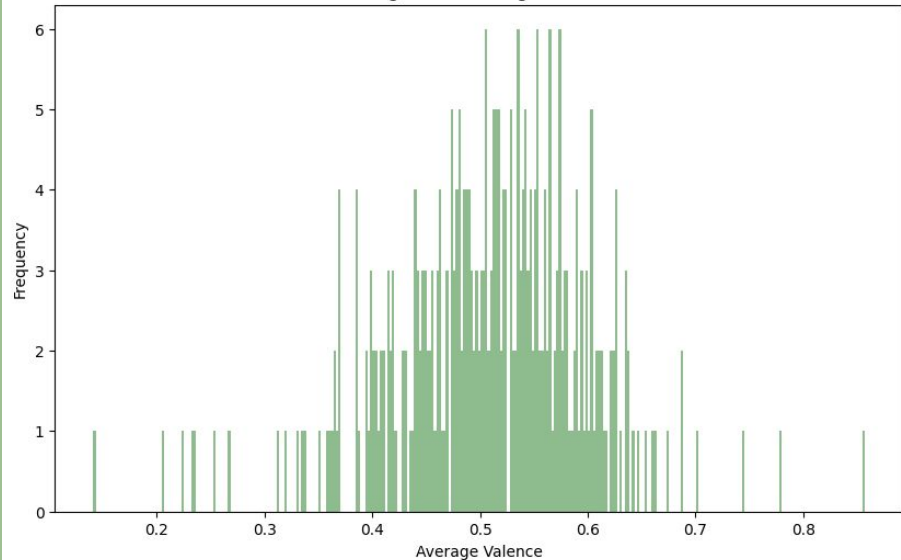


Weather

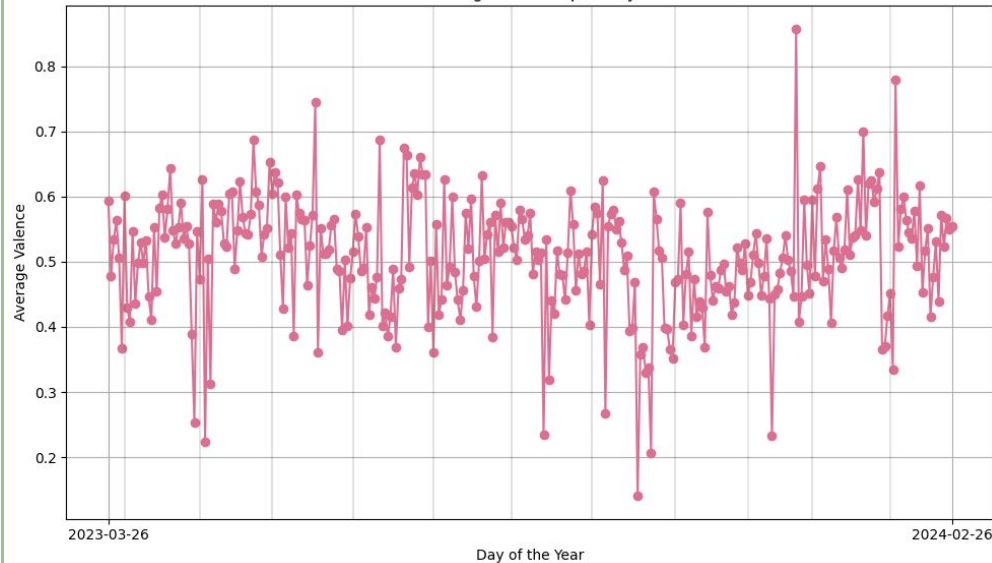
- Using the good weather attributes described earlier collect weather data for corresponding to date range from song
- All values returned without issue



Histogram of Average Valence

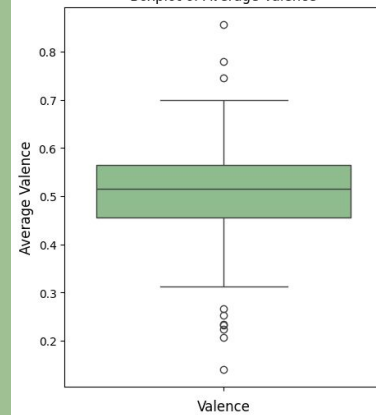


Average Valence per Day



Average Valence

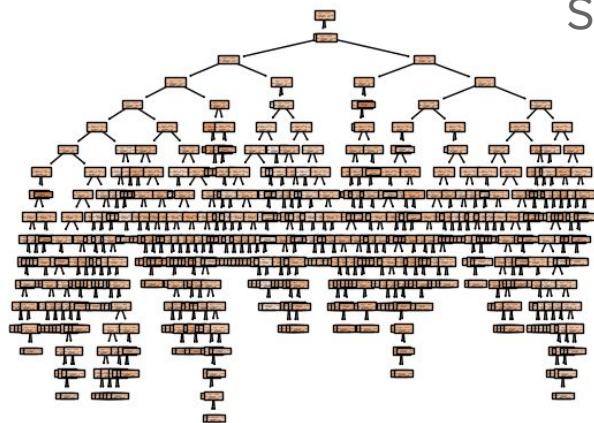
Boxplot of Average Valence



Methodology - Algorithms

scikit-learn

- Used scikit-learn LinearRegression and DecisionTreeRegressor for models.
- Fitted these models to each group member's listening history.

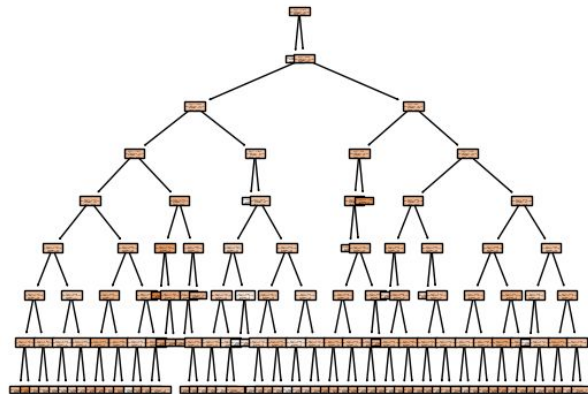


Scoring

- Used scikit-learn LearningCurve with cv=10 and negative RMSE scoring.

Model Data

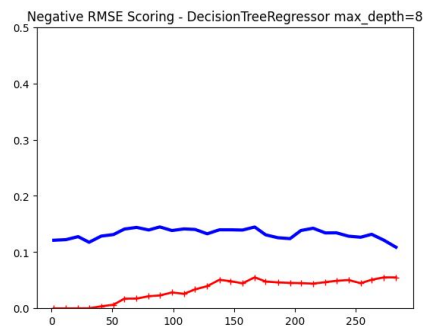
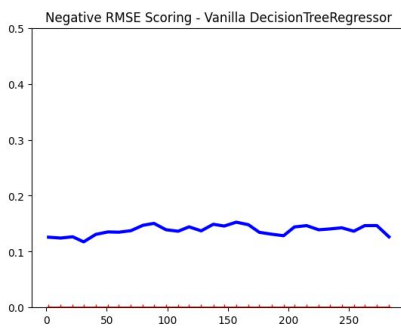
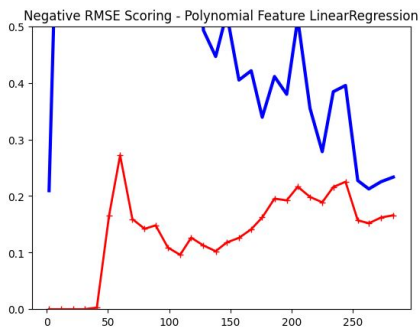
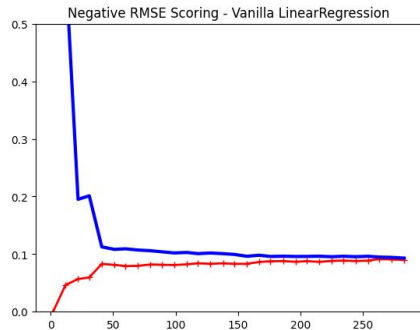
- We scaled the weather data using scikit-learn's StandardScaler for the models.
- The scaled DataFrame without the valence column was used as the features, the valence column was the label.



Preliminary Results

Linear Regression

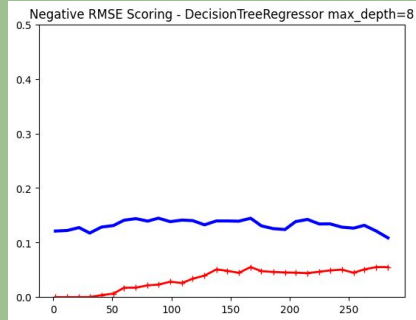
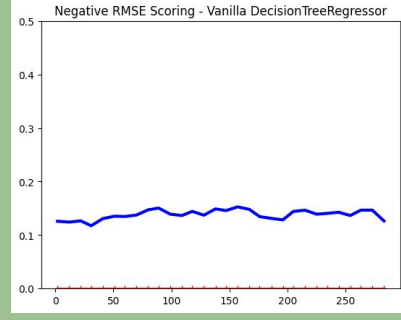
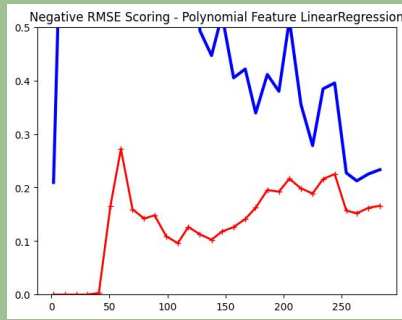
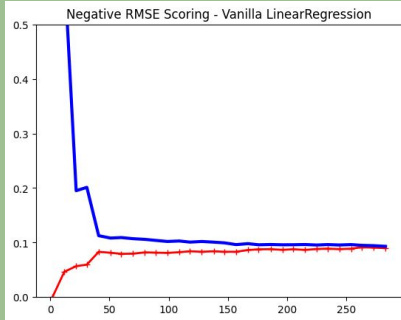
- Tested with and without poly_features
- We see the poly_features with degree 2 performs sporadically compared to vanilla, and >2 performed much worse
- Overall, the RMSE was 0.1 at best and 0.2 at worst, which is quite poor considering the standard deviation of valence by day was 0.10, 0.15, and 0.14 for the different member's histories



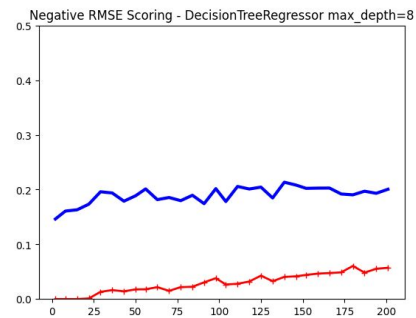
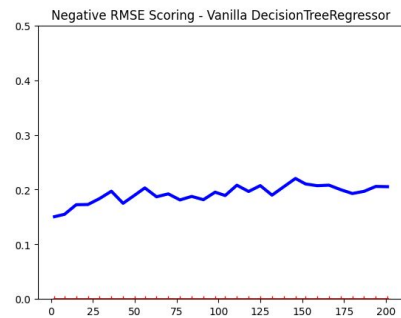
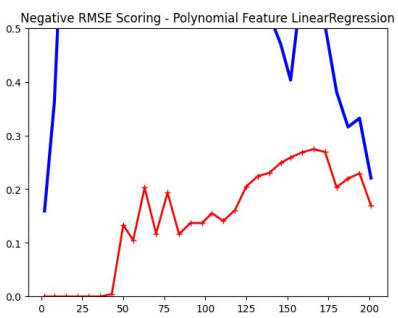
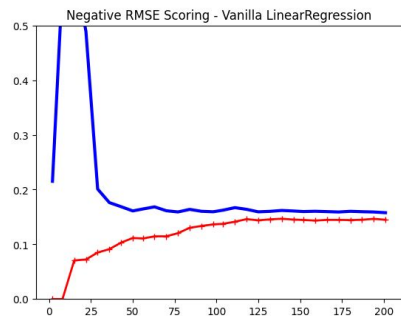
Decision Tree Regressor

- Tested with vanilla and max_depth=8
- We see the max_depth=8 has higher training error, but similar RMSE (slightly smaller)
- The RMSE was very similar to Linear Regression

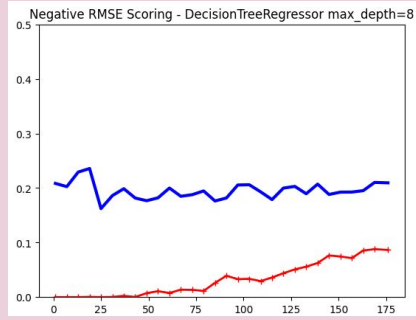
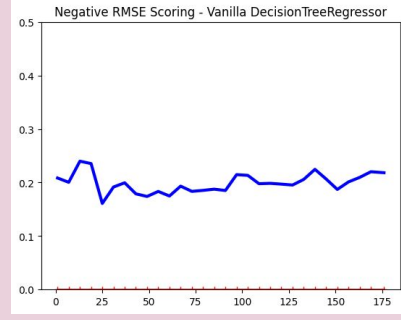
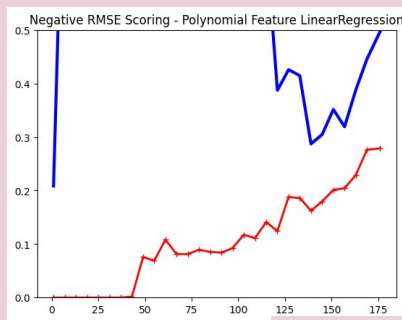
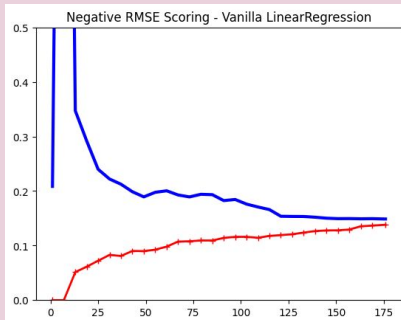
315 points



224 points



196 points



Changes depending on number of data points